

CHRISTOPHER MIKAL LOGAN

Senior Software Engineer & Architect | AI/ML Engineering | Cloud & Systems Architecture

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Location: Goshen, NY | Citizenship: U.S. Citizen | Veteran's Preference: N/A | Federal Civilian: No

PROFESSIONAL SUMMARY

Results-driven Senior Software Engineer and Architect with 30+ years of progressive experience designing, developing, and leading enterprise-scale technology solutions across financial services, defense, medical devices, and government sectors. Demonstrated expertise in Artificial Intelligence, Machine Learning, symbolic AI, hybrid reasoning systems, Retrieval-Augmented Generation (RAG), large language models (LLMs), cloud infrastructure, and full-stack application development. Proven track record reducing operational costs, modernizing legacy systems, and delivering measurable performance improvements — including a 93% reduction in batch processing time and migration of 100+ applications to CDN infrastructure offloading 95–99% of server traffic. Experienced CTO-level advisor with deep competency in regulatory compliance (NIST, FDA, SOX, NACHA), intellectual property management, and cross-functional team leadership. Holds an M.S. in Computer Science and active security-related background from defense work at Raytheon Systems and Technology Service Corporation.

CORE COMPETENCIES & TECHNICAL SKILLS

Artificial Intelligence & ML: Generative AI, Symbolic AI, Hybrid Reasoning Systems, Agentic Workflows, RAG Agents, LLM Integration, Neural Network Design, PyTorch, TensorFlow, OCR Modernization, OpenAI API, GitHub Copilot, LangChain, ChromaDB, Ollama

Cloud & DevOps: AWS (EC2, S3, CloudFront), Akamai CDN/NetStorage, Docker, CI/CD Pipelines, Jenkins, GitHub Actions, GitLab, Internal Cloud Platforms

Programming Languages: Python, Common Lisp, Java, JavaScript, TypeScript, C++, C, SQL (PL/SQL, T-SQL), PHP, FORTRAN, Shell/Bash Scripting, VB, Swift

Frameworks & Libraries: Spring Boot, Angular, AngularJS, React, Node.js, D3.js, REST APIs, SOAP/Apache Axis, WebLogic Portal, Form.io, Hazelcast, Lucene

Databases: Oracle DB / PL/SQL, DB2, MySQL, PostgreSQL, Sybase, Access 97, ChromaDB (vector store)

Systems Architecture: Scalable Cloud-Native Design, High-Availability Systems, Mainframe-to-Cloud Migration, Microservices, Embedded ARM/IoT Systems, Real-Time Data Acquisition

Finance & Payments: ACH/NACHA File Processing, Real-Time Risk Systems, Trading Platforms, Fintech API Integration

Security & Compliance: NIST Frameworks, FDA Regulatory Compliance, SOX, NACHA, USPTO Patent & Trademark Filing

Mobile Development: iOS (Swift), Android (Java), Hybrid Apps (Cue-me by Openstream), Cross-Platform HTML/CSS/JS

Leadership & Management: CTO-Level Advisory, Team Leadership (5–50+), Vendor Management, Project Management, Agile/Scrum, Product Roadmapping, Fractional C-Suite Services

PROFESSIONAL EXPERIENCE

Senior Software Engineer — Medical Devices

Burleon Tech | Remote | January 2021 – Present

Hours per week: 20 | Salary: Consulting Rate

Serve as lead software engineer and architect for an emerging medical device startup developing ARM-based embedded systems with real-time sensor integration. Responsible for the full technical scope of the device platform including hardware interfacing, software architecture, AI capability integration, regulatory compliance, and intellectual property management.

- Design and develop embedded software for ARM-based CPUs using Reduced Instruction Set Computing (RISC) architecture, implementing real-time data acquisition pipelines with GPIO pin integration for sensors, LEDs, and motors.
- Architect and integrate Artificial Intelligence and Machine Learning capabilities into medical device firmware, enhancing device intelligence and clinical utility.
- Develop real-time user interfaces for device monitoring and control, applying Python as the primary development language across firmware and application layers.
- Implement Internet of Things (IoT) connectivity, enabling device telemetry, remote monitoring, and data transmission to cloud-based platforms.
- Navigate FDA regulatory compliance requirements for medical device software, applying standards relevant to Software as a Medical Device (SaMD) and embedded health technology.
- Manage patent and intellectual property processes, collaborating with legal counsel to protect proprietary algorithms and device innovations.
- Provide technical documentation and architecture design deliverables aligned with FDA submission requirements and quality management standards.

Founder / Lead Developer / Fractional CTO

Worlwin LLC | New York, NY (Remote/Hybrid) | January 2009 – Present

Hours per week: 40 | Salary: Variable Consulting Revenue

Founded and operate an independent technology consultancy providing fractional CTO services, software architecture, full-stack development, and technology strategy to startups and established firms across multiple industries including real estate, healthcare, government, finance, and media. Manage all aspects of client engagement from requirements gathering through delivery and ongoing support.

- Created an AI-native hybrid reasoning platform that combines symbolic AI, LLM-style retrieval orchestration, and live ML-era data pipelines; engineered 7 global source adapters, normalized 27+ evaluable signals into a unified schema, and delivered an interpretable nation-scale scoring system with transparent evidence, grounded outputs, and constitutional alignment diagnostics.
- Developed an AI-enhanced Kitchen Display System (KDS) using multi-LLM orchestration, RAG, n8n, order extraction, and automated test cases, improving chatbot interaction performance by 60%.
- Provided fractional C-level technology leadership to multiple client organizations simultaneously, including strategic roadmap development, technology stack selection, vendor negotiation, and team building.
- Designed and developed full-featured property management platform for Calonex, including public website, integrated payment gateway, and native mobile application, delivering end-to-end client-facing and back-office capabilities.
- Engineered IT management system for the New York City Department of Information Technology and Telecommunications (DoITT), creating visualization and inventory tools for data center infrastructure — enabling structured management of complex government technology assets.
- Built NYASH Health platform enabling users to research medical topics, search for healthcare employment opportunities, and access detailed profiles of medical facilities, incorporating search optimization and accessibility best practices.
- Developed high-throughput mass messaging broadcast system for MediaKonnnect, enabling concurrent delivery of time-sensitive notifications and updates to large user populations.
- Implemented and maintained National Automated Clearing House Association (NACHA) file processing for ACH transactions, enabling secure bank-to-bank payment automation for payroll, vendor disbursements, and customer billing workflows.
- Developed Angular-based shared-service web application wrapping D3.js for dynamic, interactive data visualization including charts, graphs, and real-time data updates across client dashboards.
- Restructured MySQL databases, implementing master-slave clustering to achieve high availability, horizontal scalability, and client-specified uptime SLAs.
- Built PHP web tools using Smarty templating engine for client-facing portals, improving usability and guiding users through complex multi-step workflows.
- Managed end-to-end USPTO trademark registration processes for multiple clients, including branding creation, application submission, office action responses, and IP protection strategy.

- Led OCR modernization initiative leveraging neural network design with PyTorch and TensorFlow, significantly improving document recognition accuracy over legacy rule-based systems using contemporary Machine Learning techniques.
- Delivered technology research, vendor assessment, and product management services to early-stage startups, accelerating time-to-market and establishing sound architectural foundations.

Sr. Vice President / Application Manager / Sr. Principal Developer

Bank of New York Mellon Corporation (BNY Mellon) | New York, NY | February 2001 – January 2025

Hours per week: 40 | Salary: \$180,000–\$220,000 (estimated final range)

Served in progressive senior technical leadership roles over 24 years at one of the world's largest global financial institutions. Held responsibility for enterprise application management, software architecture, team leadership, Generative AI strategy, cloud infrastructure, mobile development, and production operations across multiple business domains including Alternative Investments, Client Onboarding, Reporting, and AI platforms.

- Early adopter of firm's Generative AI engineering efforts, leveraging publicly available Large Language Models (LLMs) to design and deploy a custom AI agent serving as an augmentation layer for business-specific workflows — enabling a self-service model to eliminate centralized bottlenecks. Employed Retrieval-Augmented Generation (RAG) to improve accuracy and relevance for domain-specific queries.
- Facilitated GitHub CoPilot integration across development teams and managed a strategic partnership with OpenAI to expand the firm's AI engineering capabilities and adoption.
- Led NEXEN frontend migration — a core fintech platform delivering data analytics, web applications, and APIs to internal and external bank clients — managing the full development lifecycle including architecture, development, and deployment.
- Designed and developed a firm-wide reporting system that reduced batch processing load times by 93%, from 8–16 hours down to 1 second–2 hours, directly improving business operations and SLA compliance.
- Built and deployed a centralized Content Delivery Platform using AWS and internal cloud infrastructure with Akamai NetStorage as the static asset backend. Migrated 100+ white-label applications to Akamai's CDN, offloading 95–99% of traffic from internal servers and significantly improving performance while reclaiming internal infrastructure capacity.
- Developed RESTful APIs and integrated them into Jenkins-based CI/CD pipelines to automate asset deployment, extending the CDN architecture across the full enterprise portal.
- Redesigned the n×n iframe architecture by consolidating fragmented embedded components into a single reusable module, implementing dynamic iframe resizing for responsive delivery and adding session keep-alive mechanisms to improve reliability.
- Led mobile portal infrastructure development including security architecture, content delivery, and App Store lifecycle management for iOS and Android platforms. Technologies used: Cue-me by Openstream for hybrid apps, Swift for iPhone/iPad, Java for Android, and HTML/CSS/JavaScript for cross-platform engagement.
- Created enterprise-grade Web Services using SOAP and Apache Axis to integrate all internal processes and legacy systems into distributed architectures, enabling cross-platform data exchange across mainframe, mid-tier, and modern cloud systems.
- Managed and led development teams ranging from 5 to 50+ engineers, overseeing project delivery, vendor teams, and cross-functional technology stacks as Application Manager for Oracle/PL/SQL-based applications.
- Applied technology stack spanning Java, Spring, Spring Boot, Angular, TypeScript, JavaScript, HTML, CSS, C++ with GDB, Oracle DB, DB2, Lucene, Hazelcast, REST, WebLogic Portal, Form.io, Selenium, and Chrome DevTools across multiple concurrent enterprise programs.
- Executed Sybase-to-Oracle database migration and ongoing Oracle/PL/SQL performance optimization, improving query throughput and database reliability for high-volume transaction systems.
- Managed 24/7 production incident response and on-call support operations, resolving critical platform issues for systems supporting global financial operations.

Software Engineer — Optics & Adaptive Systems

Raytheon Systems Company (RSC) | El Segundo, CA | January 1997 – January 2001

Hours per week: 40 | Salary: Approximately \$65,000–\$80,000

Contributed to advanced software engineering efforts supporting defense optics programs, adaptive optics (AO) systems, and Hubble Space Telescope data analysis. Worked across UNIX/Windows environments on projects involving real-time systems, network communications, and scientific data processing.

- Ported critical optics library package from VMS FORTRAN to DEV Visual FORTRAN, maintaining numerical accuracy and performance parity while modernizing the development environment.
- Created photomask designs for lens fabrication using PC and GDSII format with L-Edit tooling, supporting precision optical manufacturing workflows.
- Contributed to adaptive optics (AO) software development on UNIX platforms in C++, maintaining over 500,000 lines of production code as one of two senior software engineers at project completion.
- Developed Java network communication components connecting a Java client to a C-language server controlling a VxWorks Real-Time Operating System (RTOS) running on Solaris, using socket-based IPC.
- Performed Hubble Space Telescope (HST/NHST) data analysis using MSVC++, DEC FORTRAN, and UNIX toolchains, supporting scientific mission data processing.
- Automated build systems using shell scripts and Makefiles, improving build consistency and reducing integration errors across distributed development teams.
- Developed GUI applications on UNIX using standard windowing toolkits; created VB-based document parser integrated with Access 97 database using stored and dynamic queries.
- Conducted internal training courses on the adaptive optics system for engineering staff; authored technical user manuals and software documentation for deliverable systems.

Member of Technical Staff — Defense Systems

Technology Service Corporation (TSC) | Silver Spring, MD | June 1994 – December 1996

Hours per week: 40 | Salary: Approximately \$45,000–\$55,000

Contributed to radar and sensor system software development for defense applications. Engaged in full lifecycle software engineering including modeling, analysis, porting, GUI development, and technical writing for classified and unclassified defense programs.

- Performed sensor analysis and modeling using C++, MATLAB, Xpatch, and additional specialized modeling software in UNIX environments.
- Ported the Radar Imagery Generator (RIG) from VAX/VMS to SUN/Solaris platform using FORTRAN, preserving computational accuracy and operational functionality.
- Developed software for radar sensor analysis including GUI tools, analysis utilities, data visualization, graph generation, and database visualization applications using C++ on UNIX.
- Constructed model databases for sensor analysis using UNIX and USGS datasets, supporting radar cross-section modeling and signature analysis workflows.
- Authored user manuals and technical documentation for deliverable defense software; responsible for modeling targets and writing production-quality deliverable software.
- Attended X Conference in New York City; completed GUI development seminar using Motif, expanding cross-platform GUI engineering capabilities.

EDUCATION

Master of Science (M.S.) — Computer Science

Pace University | New York, NY

Bachelor of Science (B.S.) — Computer Science

Andrews University | Berrien Springs, MI

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- USPTO Trademark Registration Process — End-to-End Practitioner (via client engagements)
- GitHub Copilot Integration — Enterprise Deployment (BNY Mellon, 2023)
- OpenAI Strategic Partnership — AI Competency Development (BNY Mellon, 2023)
- Motif GUI Development Seminar — X Conference, New York City
- Adaptive Optics Systems Training — Raytheon Systems Company (Instructor)

KEY TECHNICAL ACCOMPLISHMENTS

- 93% Reduction in Batch Processing Time: Redesigned firm-wide reporting system at BNY Mellon, cutting report generation from 8–16 hours to 1 second–2 hours.
- CDN Migration — 95–99% Traffic Offload: Architected and executed migration of 100+ white-label applications to Akamai CDN, dramatically reducing internal server load and improving end-user performance.
- Generative AI / RAG Agent Deployment: Led enterprise AI agent development at BNY Mellon leveraging LLMs, RAG, and OpenAI partnership to deliver domain-specific, self-service AI capabilities.
- OCR Modernization with Neural Networks: Led PyTorch/TensorFlow-based neural network redesign of OCR pipeline, achieving significant accuracy improvements over legacy systems.
- DoITT Government Infrastructure Visualization: Engineered data center inventory and visualization system for New York City's Department of Information Technology and Telecommunications.

AI-Native Civilizational Evaluation Platform: Built The Adamas Code using Common Lisp, TypeScript, and React; integrated 7 global data adapters and 27+ normalized signals to deliver interpretable nation-scale scoring, grounded evidence, and constitutional alignment diagnostics.

- Medical Device AI Integration: Designing AI-enhanced embedded ARM medical device software with real-time sensor integration and FDA compliance at Burleon Tech.
- USPTO IP Management: Managed trademark registration and IP protection end-to-end for multiple clients through the United States Patent and Trademark Office.

ADDITIONAL INFORMATION

Availability: Available immediately for full-time federal or senior technology positions

Work Authorization: U.S. Citizen — eligible for federal employment and security clearance sponsorship

Background: Prior defense contractor experience at Raytheon Systems Company and Technology Service Corporation; familiar with controlled environment work standards

GitHub: github.com/pedalshoe

Interests: Acting (live and film), Piano performance, Photography, Drawing

— End of Resume —